Dmytro Palahin

Data Engineer

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Languages: English - **C1** · French - **C1** · Ukrainian - **native** · Russian - **bilingual**

From data migration to tooling for data teams, the same standard: rigorous validation, technical autonomy and attention to business needs.

PROFESSIONAL EXPERIENCE

Data Engineer / MLOps (Apprenticeship) **09/2023 - 08/2026**
Société Générale Insurance *La Défense, France*

- Designed a **SAS-to-Python migration system built on AI agents (GitHub Copilot)**: **5-step workflow**, specialized **Skills**, persistent context (*copilot-instructions.md*), a module replacing SAS macros and **2-level tests**
  - Python · Polars · DuckDB · pytest · GitHub Copilot · Prompt Engineering · SAS · AWS S3
- Automated a **migration pipeline for 500+ GB of SAS datasets** from SMB network shares **to AWS S3**: benchmarked 4 Python libraries, **streaming multipart upload** for files over 5 GB
  - Python · boto3 · fsspec · s3fs · smbprotocol · AWS S3 · Linux · Bash
- Built an **automated inventory of 500+ SAS programs** (SAS Enterprise Guide, GitLab): execution metadata, commit history, **dependency mapping** and a **migration priority score**, presented to managers
  - Python · Pandas · Git · openpyxl · Matplotlib · Seaborn · GitLab
- Evaluated the **Zaion Callbot** on **1 year of data**: record matching without a common identifier, **confusion matrix** (about 80% of call reasons recognized), **decision tree** on recognition errors and **3 improvement recommendations**
  - Python · Pandas · NumPy · Scikit-Learn · Matplotlib · Seaborn · Plotly · Jupyter Notebook
- Deployed an **Apache Superset dashboard for monitoring the Callbot**: **10+ operational KPIs** (customer identification, call duration, callbacks) computed with **SQL queries** on PostgreSQL JSONB data
  - Apache Superset · PostgreSQL · SQL
- Maintained **code-server** and **35+ internal extensions** (**40+ users**) through **version upgrades**: compatibility testing, **automated installation scripts**, integration of Ruff and BasedPyright (10x faster analysis than Flake8)
  - Linux · Bash · code-server · JupyterHub · VS Code · Ruff · BasedPyright · Git
- Developed **3 VS Code extensions** used daily by **30+ Data Scientists and Data Analysts**, including **real-time CPU/RAM/GPU monitoring** and the **time remaining before cleanup** of JupyterHub sessions
  - TypeScript · Python · psutil · JupyterHub API · JSON · Git · VS Code

SKILLS

- **Data Engineering:** Python, SQL, Spark/PySpark, Orchestration (Airflow, Kedro), MLflow, dbt, Kafka, Jenkins, ETL/ELT, Data Pipelines/Ingestion, Data Migration, Data Modeling, Data Warehousing, SAS, FastAPI, REST APIs
- **AI & ML:** LLMs, RAG, Generative AI, AI Agents, LangChain, LangGraph, MCP, GitHub Copilot, Prompt Engineering, NLP, Deep Learning, Scikit-Learn, XGBoost, LightGBM, TensorFlow/Keras, PyTorch, Feature Engineering, Model Evaluation
- **Cloud & Data:** AWS S3, Azure, Azure OpenAI, BigQuery, PostgreSQL, MySQL, Oracle, Polars, DuckDB, Linux, NoSQL
- **Data Quality & Analytics:** Pandas, NumPy, Data Quality, Validation, Monitoring, Reporting, Apache Superset, Tableau
- **DevOps:** Java, C/C++, TS/JS, Git (GitHub), Docker, Kubernetes, CI/CD, GitHub Actions, Bash, Automated Testing, Jira

EDUCATION

Sup Galilée School of Engineering - Diplôme d'ingénieur in Computer Science (Master's level) **2023 - 2026**
Coursework: Cloud Computing, Advanced Databases and NoSQL (MongoDB, Cassandra, Redis, Elasticsearch), Data Mining, Machine Learning and Programming for AI, NLP, Software Engineering (UML, Design Patterns), Linux System Administration, Probability and Statistics, Linear and Combinatorial Optimization (MILP), Parallel and Distributed Algorithms, Graph Algorithms

Université Sorbonne Paris Nord - Integrated Preparatory Program for Sup Galilée **2021 - 2023**
Coursework: Linear Algebra, Analysis, Topology, Differential Calculus, Probability, Scientific Computing, Algorithms, UNIX Systems

AI & ENGINEERING PROJECTS

- **Email Phishing Detection System** | *Python, Machine Learning, NLP* **2026**
  Classification of 76.7k emails (5 real datasets): NLP (TF-IDF, URL extraction), LightGBM + Naive Bayes + LSTM ensemble
- **Ant Colony Optimization for the Traveling Salesman Problem** | *C11, Metaheuristics, Dynamic Programming* **2023**
  C11 library: probabilistic model + 2-opt, validated by exact Held-Karp (optimum on 500/500 instances); 117 tests, fuzzing, CI
- **Logistics Center Placement** | *Python, Julia, CPLEX, Optimization* **2025**
  Combinatorial optimization project for placing logistics centers with MILP models solved with CPLEX and heuristics
- **Reaction Systems** | *Haskell* **2024**
  Representation of reaction systems and processes in functional programming
- **Published Research Paper** | *Mathematical Algorithms* **2021**
  Impulse noise filtering in video images — *Informatics and Mathematical Methods in Simulation*, vol. 11, no. 4
- **Anomaly Detection** | *Python, Machine Learning* **2021**
  Methods for partial data labeling

CERTIFICATIONS

- **Udemy - Jose Portilla / Pierian Training:** *Python for Data Science and Machine Learning Bootcamp*
- **Huawei AI Certifications:** *Artificial Intelligence and Applications · Artificial Intelligence Technology and Applications*

AWARDS & COMPETITIONS

- **Laureate of the Georges Besse Foundation**
- **Eco-Techno Ukraine ISEF** - 1st place in Computer Science
- **Service Internet Intelligence Collective** - 1st place in Ukraine

INTERESTS

Investing · Chess · Skiing · Tennis · Swimming · Karting | Formula 1